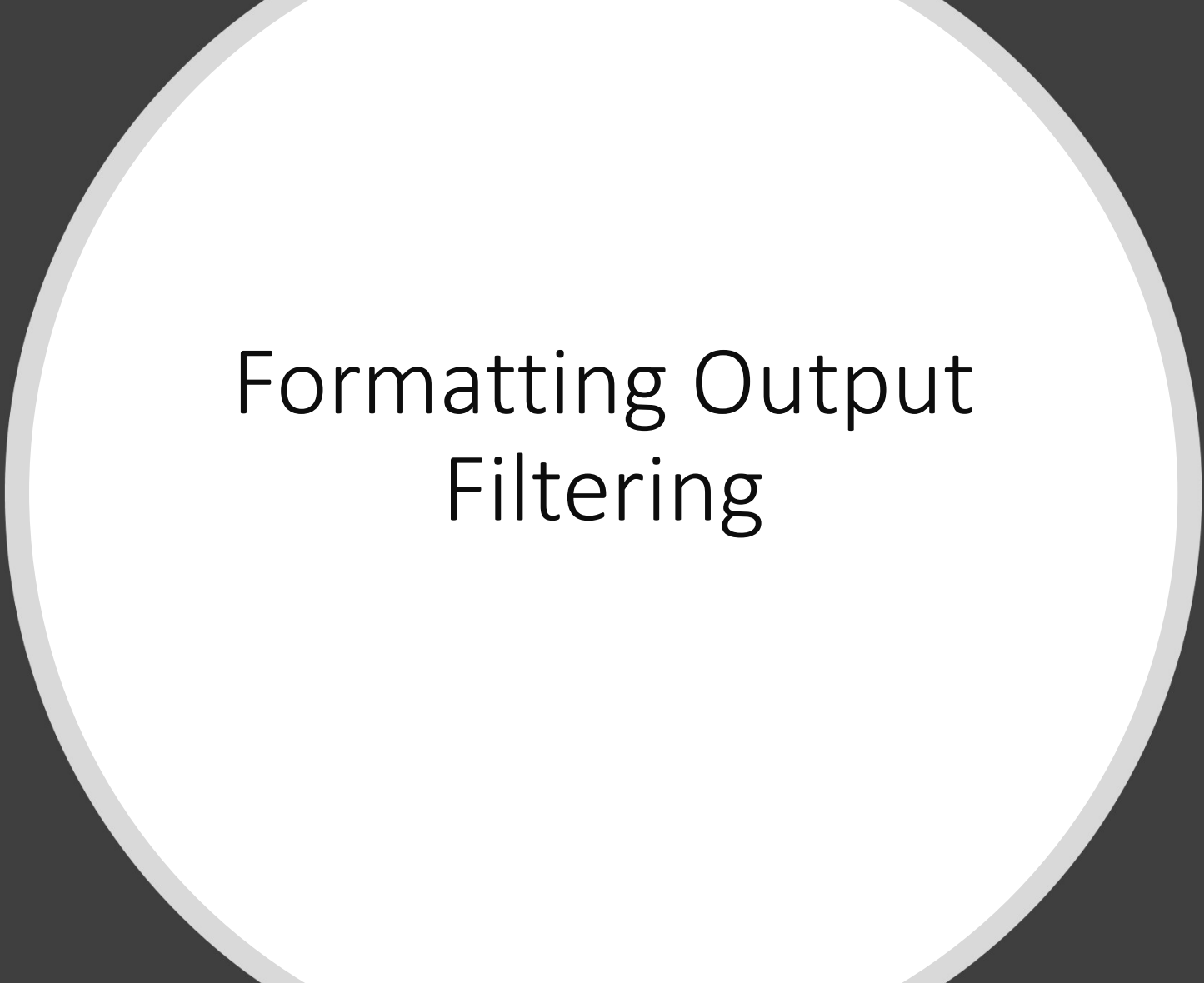




Formatting Output Filtering

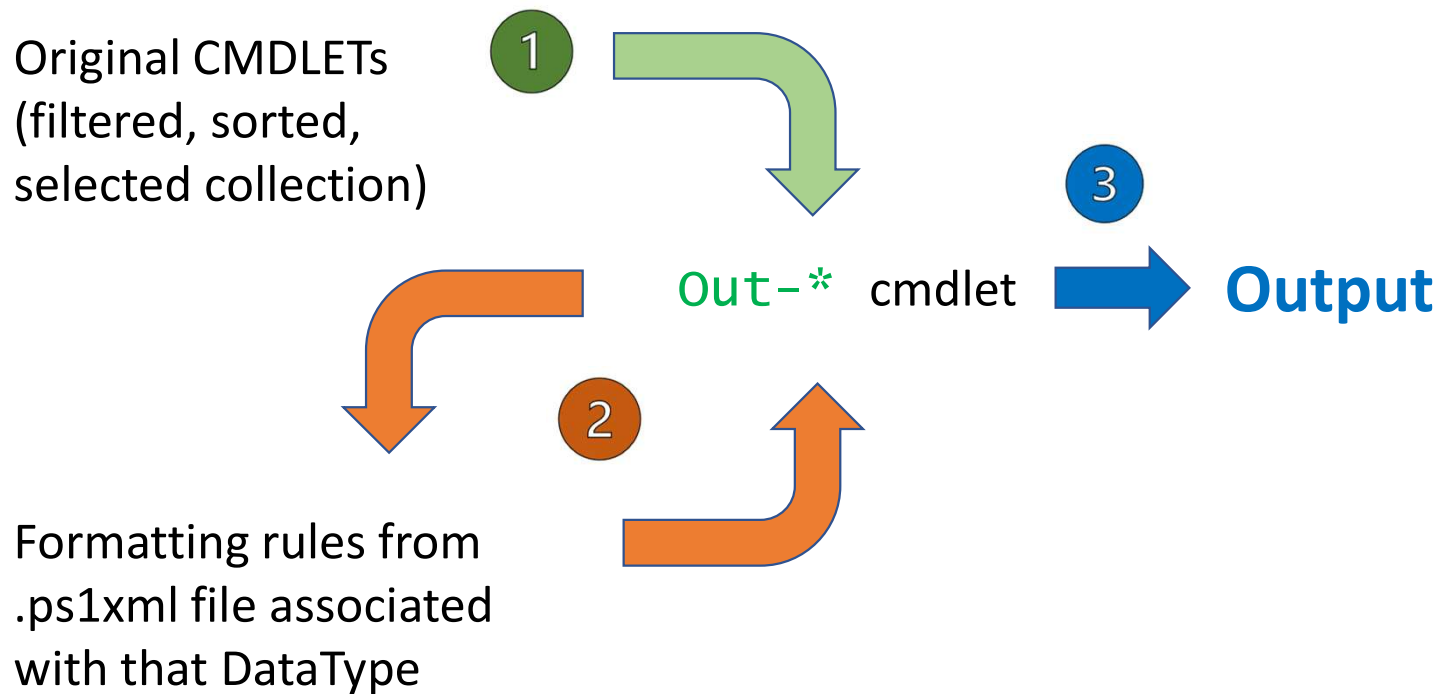
Review of Concepts

- Powershell produces **collections** as output.
- Those objects have **properties** (many of them; more than are displayed by default). We can find the types of properties by using **Get-Member**.
- If you run a command – and do nothing else – Powershell will use a default format to display the output. This default format is **built-in to PS**.
- We can change the default output, however.
 - We can use **Sort-object** to change the ordering of the rows in the collection.
 - We can use **Select-Object** to select the properties (columns) we want displayed in our table.
 - We can use **Format-** commands
 - We can use an **Out-** command to change how the object is output.
- When we do a **Select-Object**, it creates a custom table, that is kind of ugly, because we have told PS not to use the default output formatting.

How Powershell formats Output (by Default)

- Every command has a predefined set of rules to format by.
- The formatting rules are stored with the command's modules in a `.format.ps1xml` files. For example: formatting for `*-Process` commands are store in `DotNetTypes.format.ps1xml`
- Every command does an `Out-Default` after being executed
- If no other `Out-*` command is entered, `Out-Host` will be used (the 'default out' is the host).
 - If the data in the pipeline is a string → Directly to `Out-Host`
 - If the data is not a string → determine if there is a view for this object.
- The `Out-*` command passes the data to the formatting process, applies the formatting view specified for the module (which follows *three rules*), and the data is passed back to the `Out-*` command.

Formatting Flow



Rules of Formatting

(ordered steps used during the formatting process)

- **Step 1:** Attempt to format using **Objects predefined DataType view**
 - If the object is listed in a `.format.ps1xml` formatting file, it will be formatted based on the predefined format rules in this file (like `DotNetTypes.format.ps1xml` for `*-Process` commands).
- **Step 2:** Use `types.ps1xml` for custom formatting
 - If there is a `.types.ps1xml` file for this command specifically, use it.
 - If the object isn't listed in a `.type.ps1xml`, use **all** of the object's properties and make an ugly table.
- **Step 3:** Choose to display in **list**, or **table** format.
 - If the output is displaying **4 or fewer** properties, display a table.
 - If the output is displaying **5 or more** properties, use a list.

Breaking the rules

- We can override the formatting on any table using custom formatting cmdlets.
- There are four formatting commands we can use to change the formatting of our output.
 - `Format-Table`
 - `Format-List`
 - `Format-Wide`
 - `Out-GridView`



Format-Table (FT)

Useful Parameters

- **-autoSize**
 - Normally, PS tries to fit the console windows on the screen. It will fill the screen with the largest table possible (except if there is a predefined view).
 - If there are only 3 columns PS will try to stretch them across the screen.
 - **-autoSize** make PS fit the columns to their contents instead of the console width.
- **-property**
 - You can select the properties to display in the table. If you do, it will list the *real* name of the property instead of the column header stored in the formatting file.
- **-groupBy**
 - Separates the returned objects into different tables (one after each other) based on the property you've grouped by
- **-wrap**
 - Wraps the contents of a value to the width of the column. Adds a '...' after the content if it would be larger than the column.

Format-List (FL)

- Sometimes there is too much information to display for a Table to be practical.
- **FL** can be useful for displaying available properties for an object (like **GM**), but it also shows the *values* of those properties.

Useful parameter

- **-Property** (**-property ***)
 - Selects the properties to be shown in the list.
- **-groupBy**
 - Separates the returned objects into different lists (one after each other) based on the property you've grouped by

Format-wide (FW)

- **FW** displays “wide lists”.
 - Can only display one property at a time in a Wide format
 - Uses the Name property by default
 - Defaults to two columns.
- Useful Parameter
- **-property**
 - Selects the property to display (default is Name)
 - Doesn't accept wildcards
- **-col**
 - Determine how many columns to display (default 2)

Controlling Output

- **Out-Default** is on the end of every command. It applies the default formatting rules found in PS and then sends the output to an **Out-Host**
- **Out-Printer** sends output to a print device.
- **Out-File** sends output to a text file.
- **Out-GridView** (not really a formatting command) sends the output of a command to table Window.
 - It completely bypasses formatting.
 - It doesn't attempt to format the output at all, but the resulting table can be ordered or formatted like any other table in Windows.
 - The resulting table can be filtered using the GUI it provides.

Tips for using **Format-** Commands

- Always use them *last* in a statement (*format right*)
 - Remember that the **Out-** commands *consume* the formatting instructions from the previous command.
 - There is always an **Out-Default** at the end of every command, unless specified otherwise.
- Example:
 - If you're Going to use **Out-GridView**, format the table before piping to it.
`Gps | Format-Table -Property Name,ID,CPU | Out-GridView`
- You can't put more than one thing in the pipeline if your going to format it.
 - Consider **Get-Process; Get-ChildItem**
 - We can send two commands to the shell (not piped but running serially) using an “;”

Points to remember

- **Filtering** should be done first (either `-filter` parameter or `where-object`). *Filter left*.
- `Sort-Object` should be used before `Select-Object`.
- `Select-Object` should be used after `Sort-Object` but before `Format-` commands, as it can change the DataType in the pipeline.
- Always `Format-` to the right of your command (right before your standard out is processed). *Format Right*.
- Don't format multiple data types (more than one command being run).

The order of a statement

Filter → Sort → Select → Format → Output